DSCI510 FINAL PROJECT

The Analysis of U.S. Economic Stress and Social Media Seeking (2019–2024)

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INTRODUCTION

1.Research Objective

To examine how the COVID-19 **economic shock influenced individuals' behavioral preferences** on **social media** from **2019 to 2024**, specifically whether people tend to shift toward **job-seeking (functional usage)** or **entertainment (recreational usage)** under rising unemployment.

2.Research Questions

- **Q1:** What are the trends in unemployment before, during, and after COVID-19?
- **Q2:** Does higher unemployment lead to increased functional social media usage?
- **Q3:** Does higher unemployment lead to increased recreational social media usage?

3.Analytical Focus

- Unemployment trends across high-stress states
- Functional vs. recreational social media responses to economic stress

Data Sources & Approach

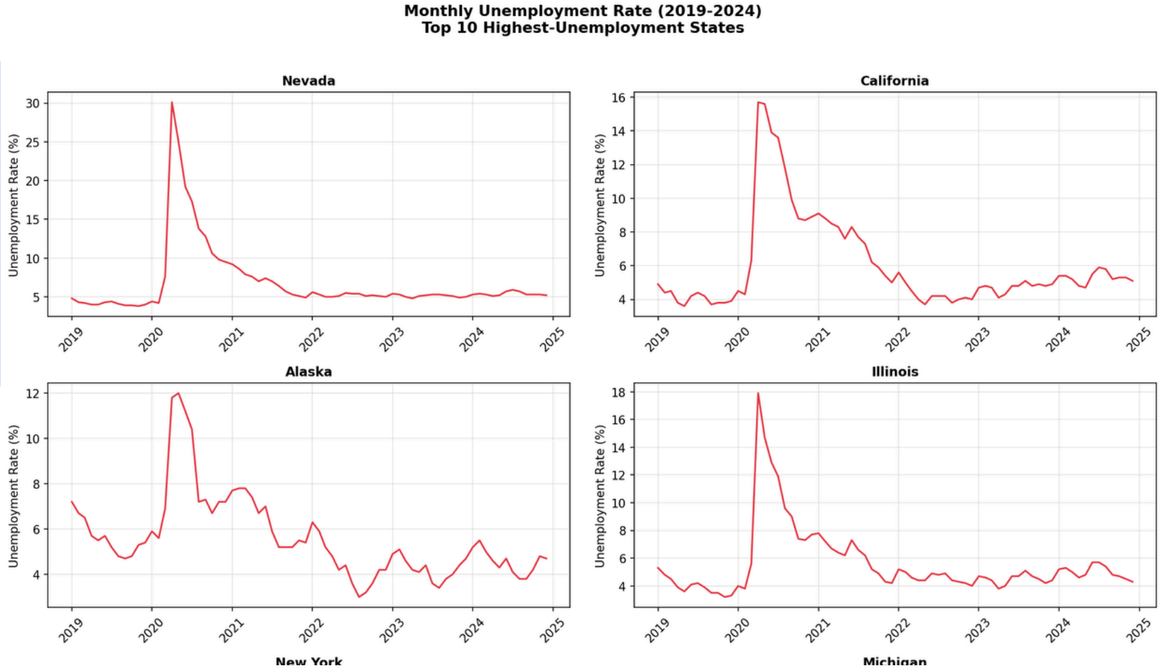
Data Source	Description	Processing Approach
BLS Unemployment Data https://download.bls.gov/pub/time.series/la/	Monthly unemployment rate by state (2019–2024). Used for measuring economic stress	Retrieved via **BLS API**; cleaned and structured as time-series data
Google Trends (Functional Social media behaviour) https://trends.google.com/trends/explore	Search interest index (0–100) for **Job-seeking Keywords Category:** “LinkedIn”, “Job”, “Hiring” to **capture job-seeking behavior**	Collected via **pytrends**; keywords **combined and averaged to construct a recreational index**
Google Trends (Recreational Social media behaviour) https://trends.google.com/trends/explore	Search interest index (0–100) for **Entertainment Keywords Category**: “TikTok”, “Netflix”, “Instagram”to **capture entertainment behavior**	Collected via **pytrends**; keywords **combined and averaged to construct a recreational index**

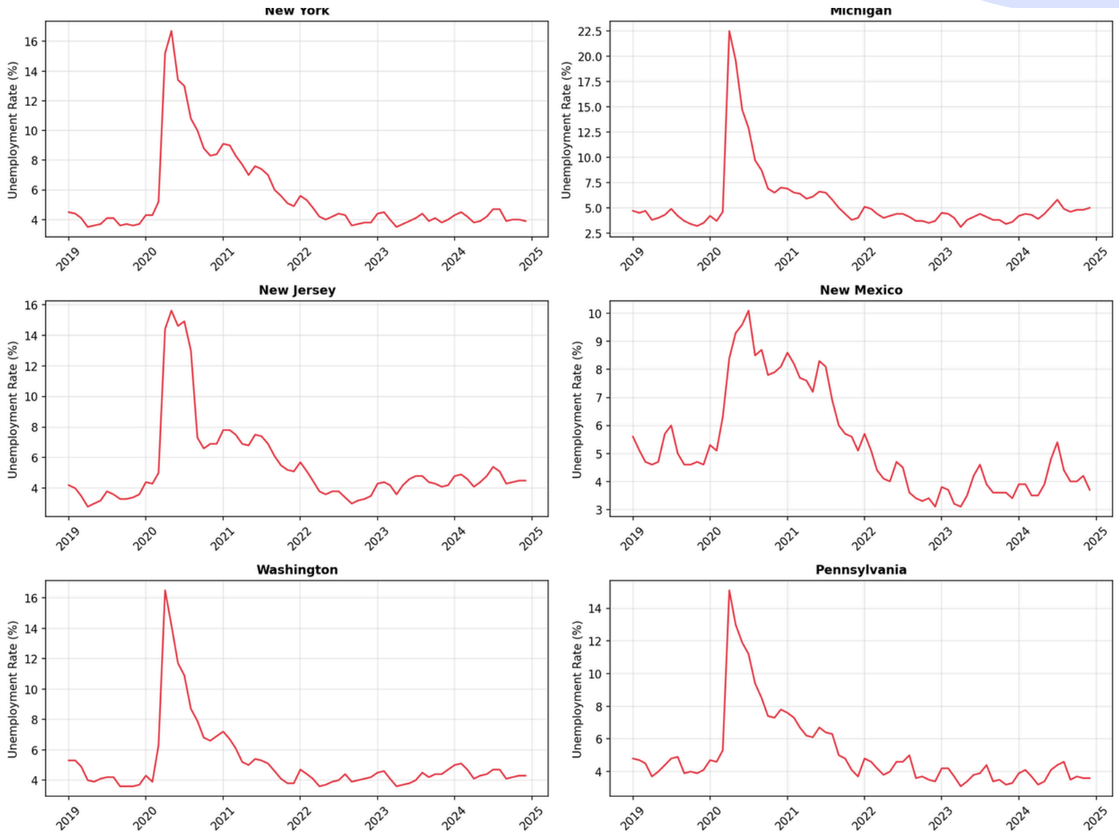
Notes:
Search interest = how popular a keyword is over time (0–100), not how many people search it, not actual search counts
Pytrends is a Python library for accessing Google Trends data, which is used to retrieve keyword-level search indices

Result 1-The unemployment trends

Unemployment peaked in 2020 and declined afterward：

- Top 10 high-unemployment states(2019-2024) identified: **Nevada, California, Alaska, Illinois, New York, Michigan, New Jersey, New Mexico, Washington, Pennsylvania**
- **Sharp increase during COVID-19 (2020)**
- **Gradual recovery** from 2021 to 2024
- Persistent variation across states

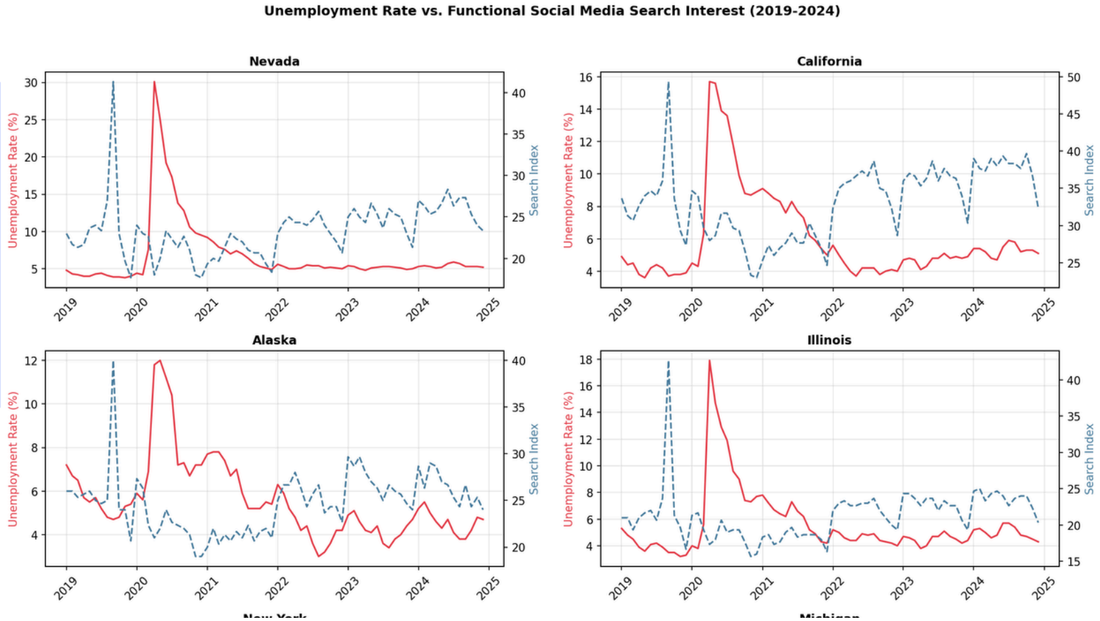


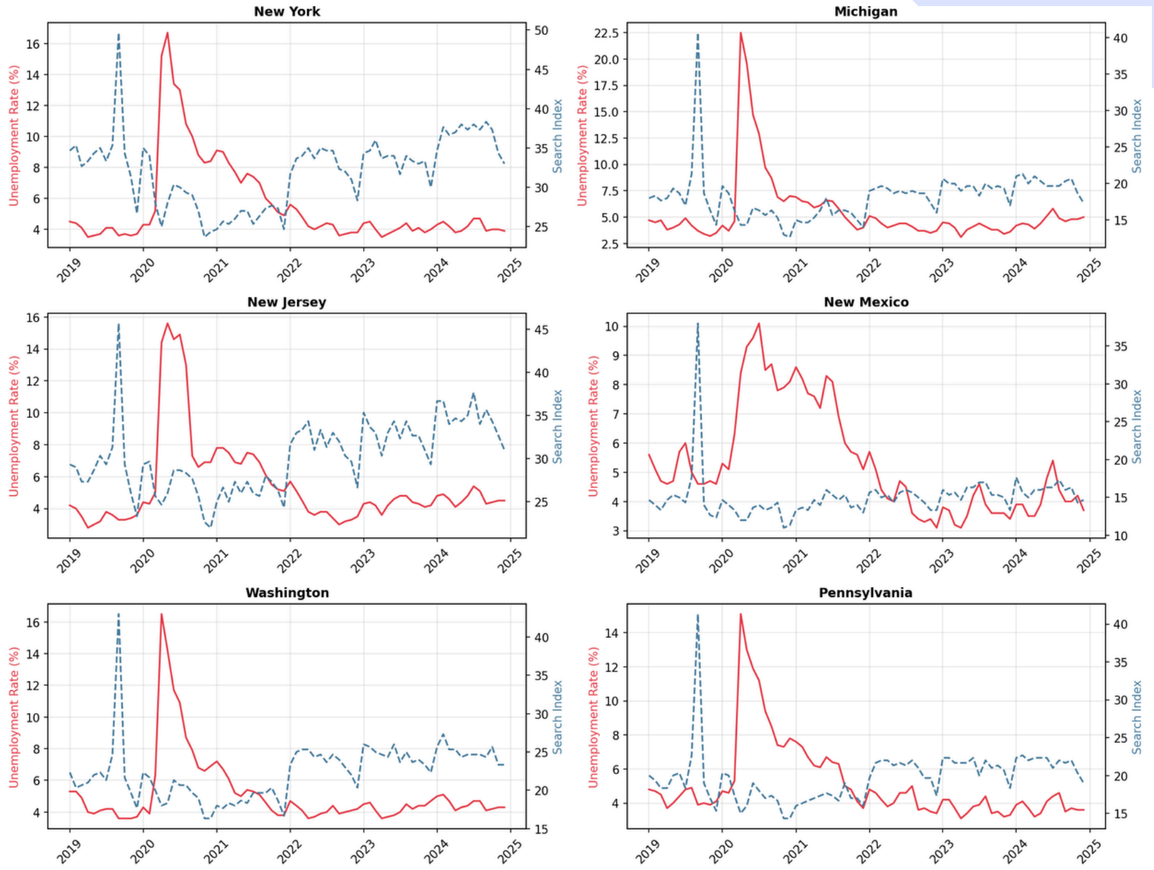


Results 2: Functional Social Media

Functional search interest increases before unemployment peaks with unemployment

- **Drops or remains low** at peak unemployment
- **Increases** after unemployment starts to **decline**
- Suggests lagged job-seeking behavior
- Indicates **delayed response to economic shock**

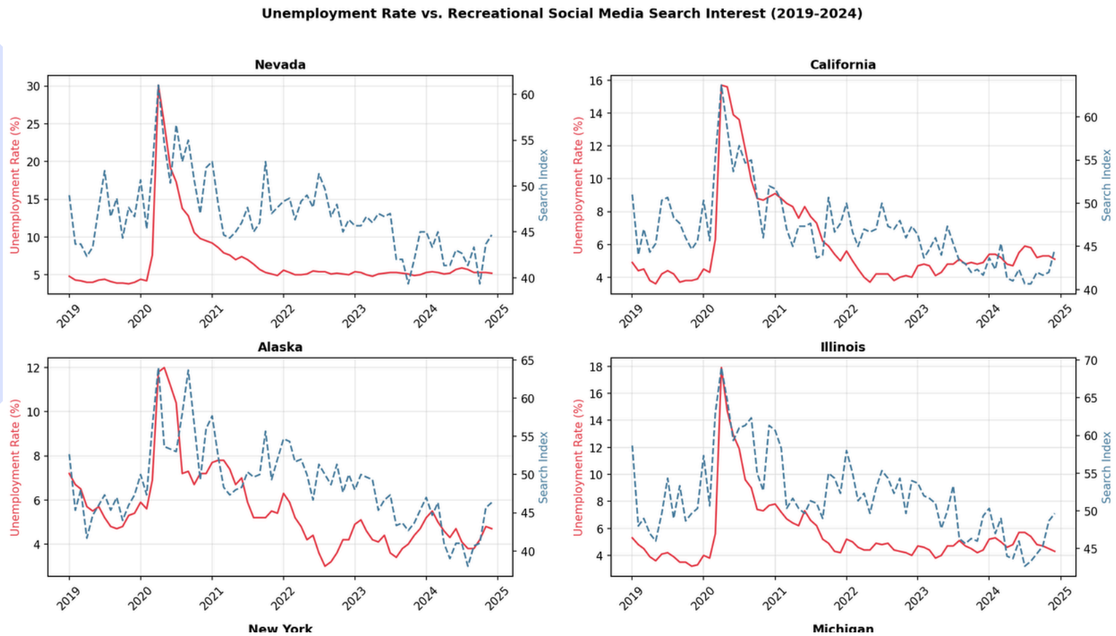


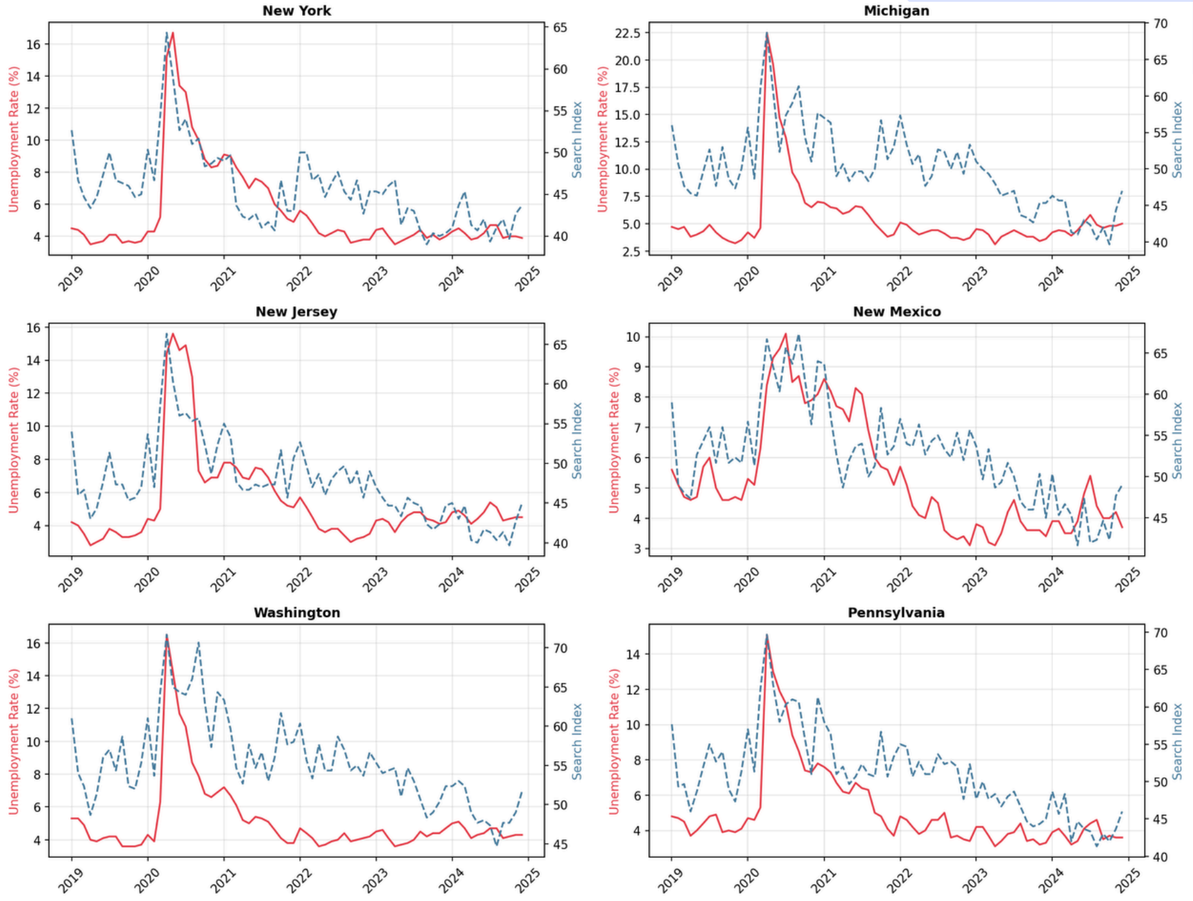


Results 3: Recreational Social Media

Recreational search interest rises during the shock but declines as unemployment recovers

- **Increases** during the 2020 unemployment spike
- **Gradually declines** as unemployment decreases
- Returns closer to pre-pandemic levels
- Suggests **a short-term response to economic shock**



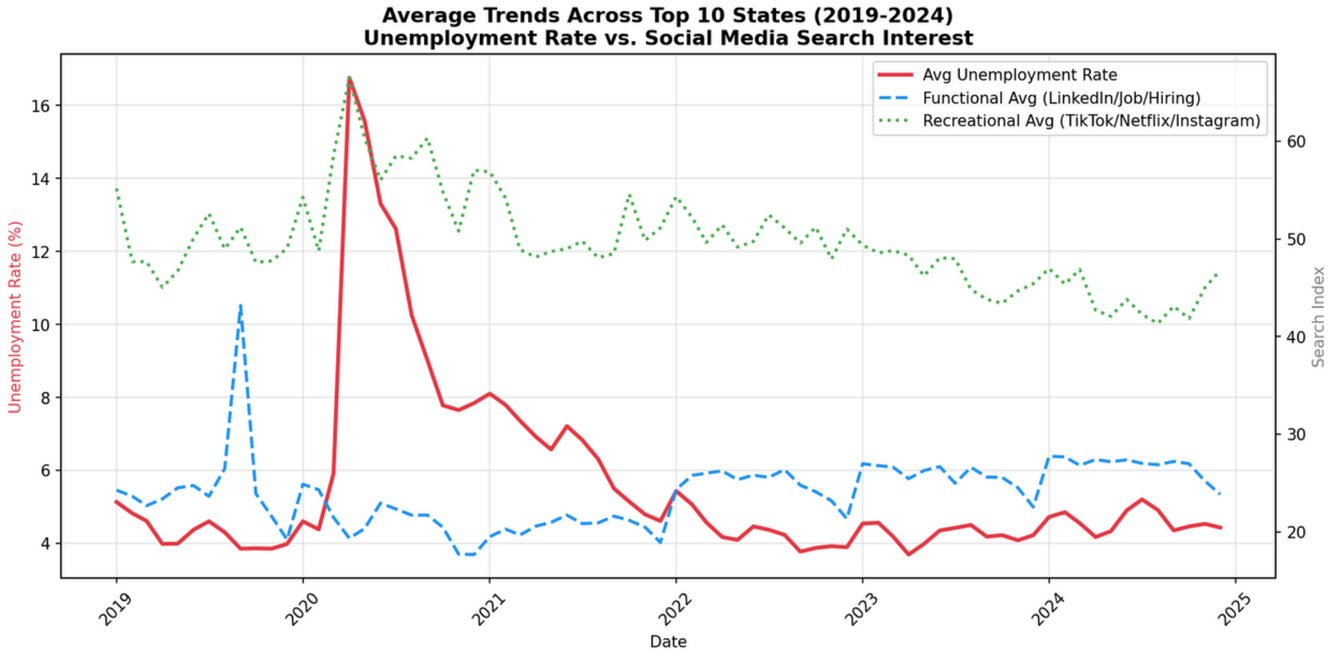


Results 4-Overall Trends

Functional and recreational Social Media behaviors show different long-term patterns

- Job-Seeking Social media behavior increases around the unemployment peak and after the unemployment peak, suggesting a growing response to economic stress.
- Entertainment Social media behavior rises during the unemployment spike and declines afterward (immediate response)

Higher unemployment → more recreational social media (Moderate positive correlation: r = 0.661, p<0.05)
Higher unemployment →less job-seeking behavior (Weak positive correlation: r = -0.400, p<0.05).

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--- Overall Summary ---
  Avg Functional correlation:   r=-0.400
  Avg Recreational correlation: r=0.661

Correlation results table:
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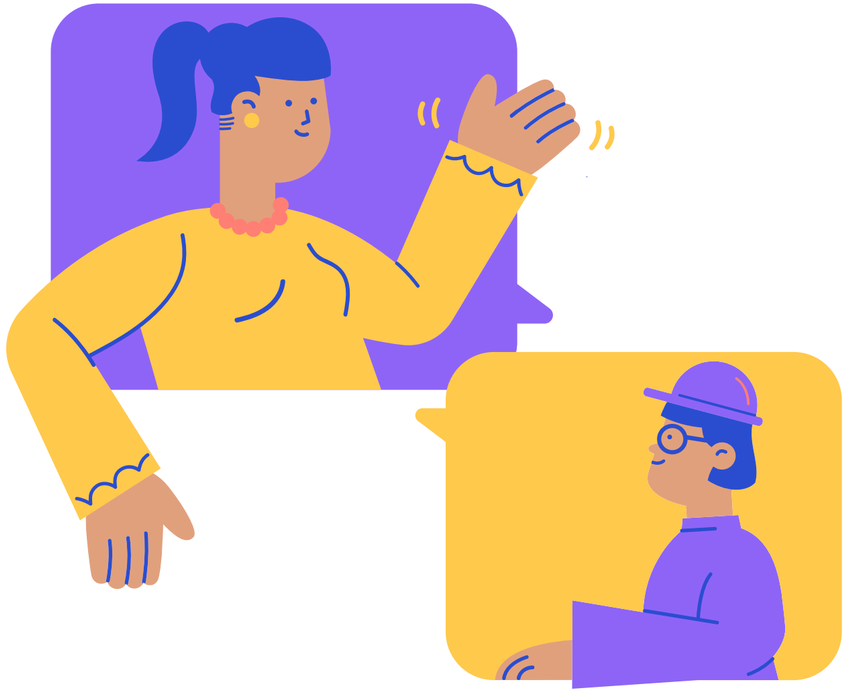
state	func_corr	func_pvalue	rec_corr	rec_pvalue
Alaska	-0.410	0.0003	0.580	0.0
California	-0.504	0.0000	0.632	0.0
Illinois	-0.365	0.0016	0.676	0.0
Michigan	-0.350	0.0026	0.557	0.0
Nevada	-0.344	0.0031	0.678	0.0
New Jersey	-0.363	0.0017	0.676	0.0
New Mexico	-0.270	0.0219	0.709	0.0
New York	-0.598	0.0000	0.663	0.0
Pennsylvania	-0.417	0.0003	0.739	0.0
Washington	-0.379	0.0010	0.704	0.0

All statistically significant

Challenge



Data Collection Challenges

- Google Trends (pytrends) may trigger **Error 429 (Too Many Requests)**
- Caused by sending frequent automated requests
- Resolved by adding time delays between requests
- **BLS API may be unstable,** returning inconsistent data sizes

Data Limitations

- Search interest is a relative index (0–100), not actual search counts
- Limits direct comparison across different keywords

Thanks for Listening